Lecture 5

Special Recurrence - Case Study

Case study

- Give an asymptotic tight bound for the following recurrence:

- $T(n) = \sqrt{n}\,T(\sqrt{n}) + n.$

How many nodes do we have in each layer?

- First layer, we have $n^{\frac{1}{2}}$ nodes.
- Second layer, there are $n^{\frac{1}{2}} * n^{\frac{1}{4}} = n^{\frac{3}{4}}$ nodes
- Third layer, there are $n^{\frac{3}{4}} * n^{\frac{1}{8}} = n^{\frac{7}{8}}$ nodes
- Layer i has $n^{1-\frac{1}{2^i}}$ nodes.

What is the cost at each layer?

- First layer has $n^{\frac{1}{2}}$ nodes, each with a cost of $n^{\frac{1}{2}}$. Total: n

- Second layer, there are $n^{\frac{1}{2}} * n^{\frac{1}{4}} = n^{\frac{3}{4}}$ nodes, each with a cost of $n^{\frac{1}{4}}$. Total: n.

- Third layer, there are $n^{\frac{3}{4}} * n^{\frac{1}{8}} = n^{\frac{7}{8}}$ nodes, each with a cost of $n^{\frac{1}{8}}$. Total: n.

- Layer i has $n^{1-\frac{1}{2^i}}$ nodes, each with a cost of $n^{\frac{1}{2^i}}$.

How many layers do we have?

- The subproblem size decreases by a square root for each increase in depth.
- Let $n = 2$ be the base case (we stop recursion when we do not have any more perfect square roots).
- Stop when $n^{\frac{1}{2^i}} = 2$ (this means we reached the last layer).

How many layers we have?

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- $n^{\frac{1}{2^i}} = 2.$
- $\log n^{\frac{1}{2^i}} = \log 2$
- $\frac{1}{2^i} \log n = 1$
- $\log n = 2^i$
- $i = \log \log n$

- The recursion tree has $\log \log n$ layers.

What is the asymptotic bound?

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- Since we have $\log \log n$ layers, each with a cost of n.
- The last layer has $n^{1-\frac{1}{2^{\log \log n}}}$ nodes.
- The cost of the internal layers is $\Theta(\text{n} \log \log n)$.
- The total cost is $\Theta(\text{n} \log \log n)$. The cost is dominated by the internal nodes of the tree.

This can be verified by substitution.

$$
\begin{aligned}
T(n) &= \sqrt{n}\,T(\sqrt{n}) + n \\
&= \sqrt{n}(\sqrt{n}\lg\lg\sqrt{n}) + n \\
&= n\lg\lg n^{1/2} + n \\
&= n\lg((1/2)\lg n) + n \\
&= n(\lg(1/2) + \lg\lg n) + n \\
&= -n + n\lg\lg n + n \\
&= n\lg\lg n \,.
\end{aligned}
$$